Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

Title: Creating Power BI Visualizations to Analyze Sales Performance Across Products, Categories, and Regions

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Creating Power BI Visualizations to Analyze Sales Performance Across Products, Categories, and Regions

GitHub Repository Link: *https://github.com/anuskaGHS/Business-Analytics*

Problem Statement

Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.
Dataset used: Sample Superstore
Source: https://www.kaggle.com/datasets/naveenkumar20bps1137/sample-superstore

Methodology / Steps Followed

Step 1: Data Collection

The Sample Superstore dataset was used for this experiment. The dataset was sourced from Kaggle and downloaded as a single CSV file. It contains retail transactional data with the following key fields:

- Order ID, Order Date, Ship Date, Ship Mode
- Customer ID, Segment, Country, City, State, Region
- Product ID, Category, Sub-Category, Product Name
- Sales, Quantity, Discount, Profit

Total Records: 9,994 rows
Total Columns: 18 (after removing Row ID)

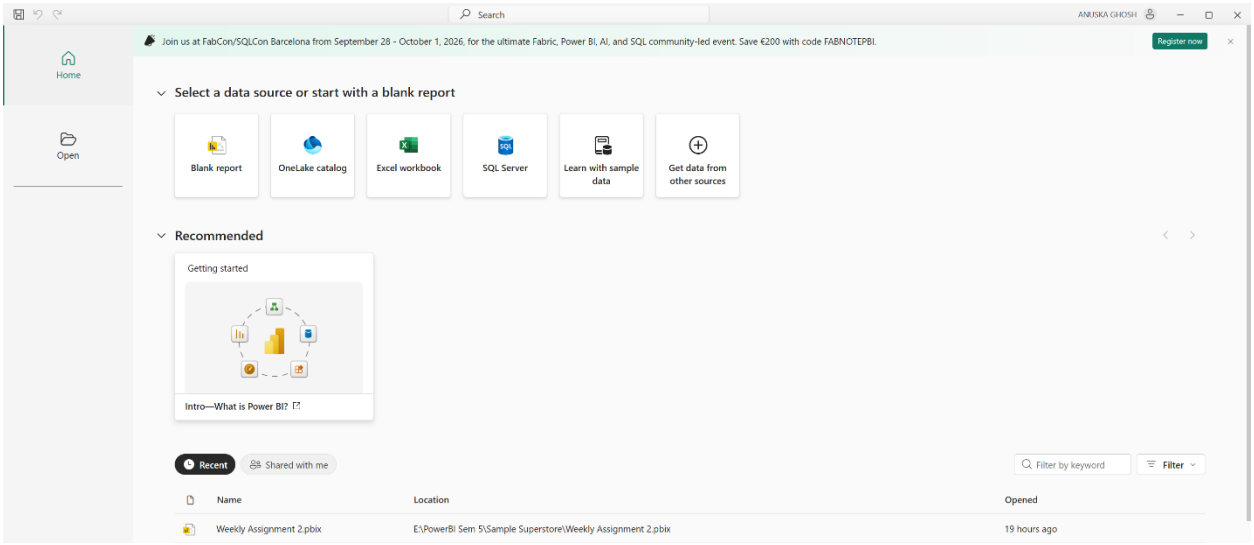


Figure 1: Power BI Desktop Home Screen

Step 2: Data Import and Transformation in Power BI

Power BI Desktop was opened and the dataset was imported using the Get Data option by selecting Text/CSV connector. The file Sample_Superstore.csv was loaded directly as a flat file.

The following transformations were applied in Power Query Editor:

- **Row ID** column was removed as it held no analytical value
- **Order Date** data type was changed from Text to Date
- **Ship Date** data type was changed from Text to Date
- **Sales, Discount,** and **Profit** data type was changed to Fixed Decimal Number

After applying all transformations, Close & Apply was clicked to load the clean data into Power BI.

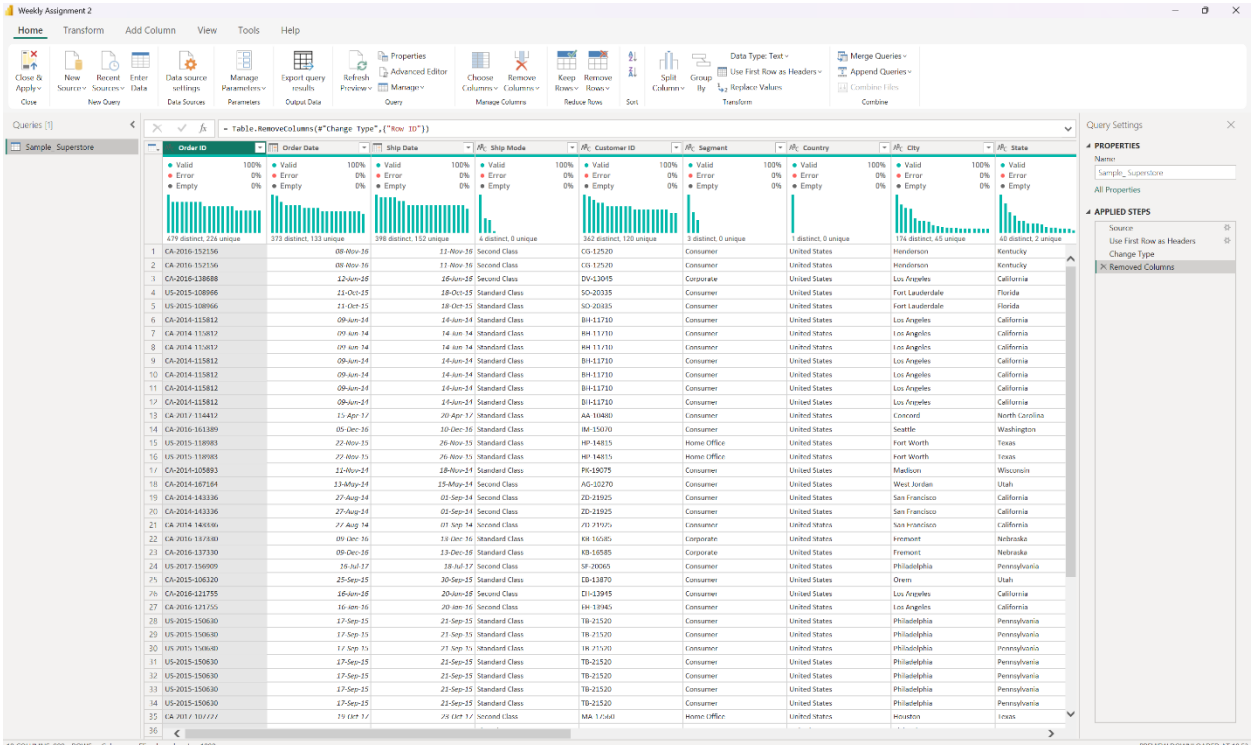


Figure 2: Power Query Editor

Step 3: Dashboard Development

An interactive dashboard titled "Superstore Sales Performance Dashboard" was developed in the Report View.
The following visuals were created:

- **3 KPI Cards -** Total Sales ($2.30M), Total Profit ($286.40K), Total Orders (5.01K)
- **Sales by Category -** Clustered Bar Chart showing sales distribution across Technology, Furniture, and Office Supplies

- **Sales Trend Over Time -** Line Chart showing yearly sales growth from 2014 to 2017
- **Sales by Segment -** Donut Chart showing sales distribution across Consumer, Corporate, and Home Office segments
- **Sales by State -** Bubble Map showing geographical distribution of sales across US states
- **Sales by Sub-Category -** Treemap showing sales breakdown within each category
- **2 Slicers -** Region and Category for interactive filtering

Output

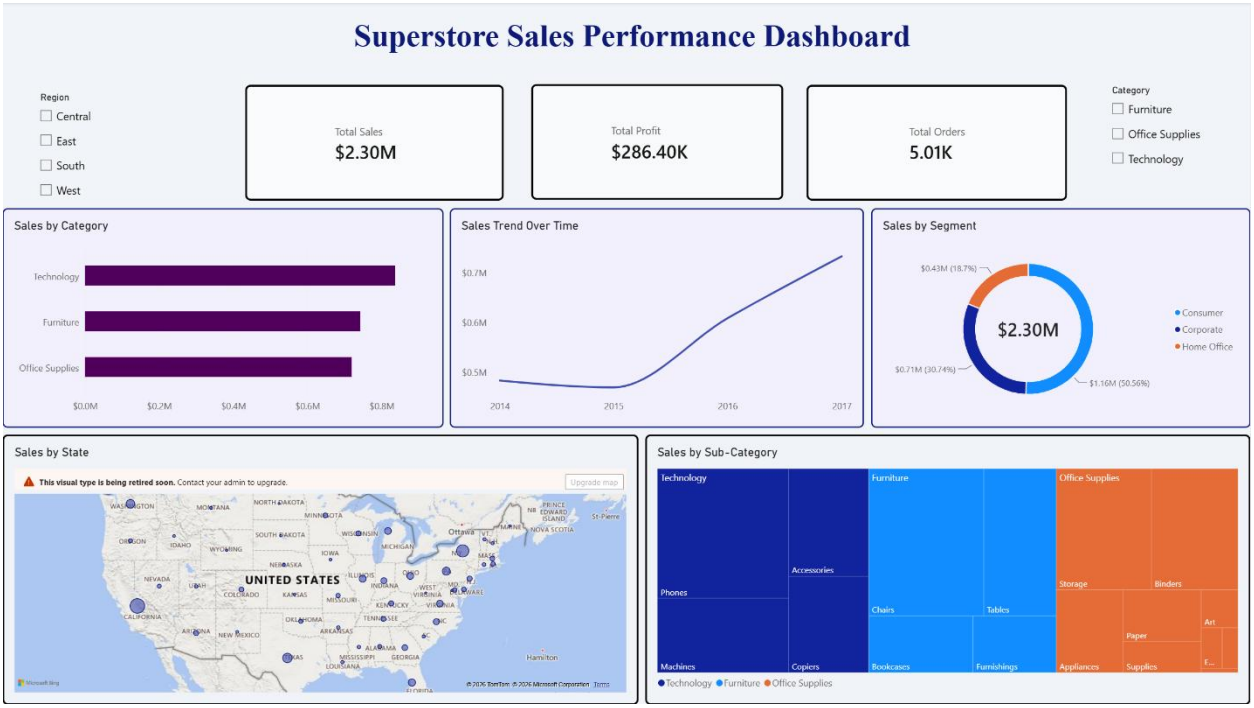


Figure 3: Superstore Sales Performance Dashboard — Report View

The final Power BI dashboard titled "Superstore Sales Performance Dashboard" successfully visualizes the retail company's business data. The dashboard provides:

- **Total Sales:** $2.30M in total revenue across all regions and categories
- **Total Profit:** $286.40K in net profit generated
- **Total Orders:** 5.01K unique orders placed
- **Category Performance:** Technology leads in sales ($0.84M), followed by Furniture ($0.74M) and Office Supplies ($0.72M)
- **Sub-Category Insights:** Phones ($330.01K) is the top performing sub-categories in Technology; Chairs ($328.45K) lead in Furniture; Storage ($223.84K) is the top performers in Office Supplies

- **Customer Segments:** Consumer segment contributes the highest sales at 50.56% ($1.16M), followed by Corporate at 30.74% ($0.71M) and Home Office at 18.7% ($0.43M)
- **Sales Growth:** Consistent upward trend in sales observed from 2015 to 2017
- **Geographic Distribution:** California shows the highest sales concentration among all US states
- **Interactive Filters:** Region and Category slicers enable dynamic data exploration across all visuals

Conclusion

This experiment successfully demonstrated the use of Microsoft Power BI to analyze retail sales performance using the Sample Superstore dataset. The dataset containing 9,994 records was imported directly from a CSV flat file, cleaned and transformed in Power Query Editor, and used to develop an interactive business dashboard.

The visualizations created including KPI Cards, Bar Chart, Line Chart, Donut Chart, Bubble Map, and Treemap, collectively provide management with a comprehensive and quick understanding of sales performance across products, categories, and regions.The interactive slicers further enable dynamic filtering for deeper business insights.

The dashboard clearly identifies key business patterns such as Technology being the top performing category, Consumer segment dominating sales, and a consistent upward sales trend from 2015 to 2017, fulfilling the objectives outlined in the problem statement.

Business Insights

The following key business insights were derived from the Superstore Sales Performance Dashboard:

- **Top Performing Category:** Technology leads in sales with $0.84M revenue, with Phones being the highest selling sub-category, indicating strong demand for tech products.
- **Customer Segmentation:** Consumer segment accounts for 50.56% of total sales, making it the most valuable customer segment for the business.
- **Sales Growth Trend:** Sales show a consistent upward trend from 2015 to 2017, with the steepest growth observed between 2015 and 2016, indicating strong business expansion.
- **Geographic Performance:** California is the strongest performing state, suggesting opportunities to expand presence in underperforming regions such as Central and South.
- **Order Volume:** With 5.01K unique orders and $2.30M in total sales, the business demonstrates strong transactional activity across the recorded period.
- **Interactive Analysis:** The Region and Category slicers enable management to drill down into specific territories and product categories for targeted business decisions.